VU VRIJE UNIVERSITEIT AMSTERDAM
Ivana Plese

Amsterdam UMC
Dr. Benno Kruit (supervisor)

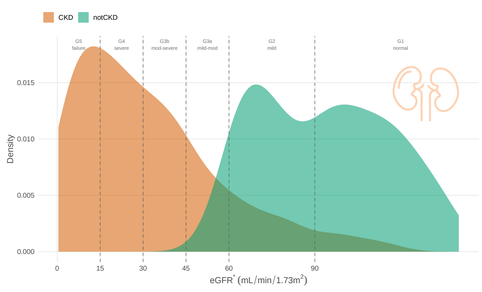
Read more and connect!

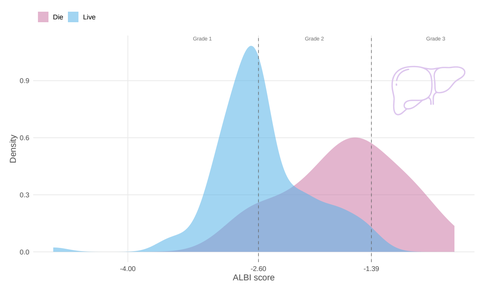
BNAIC
BeNeLearn
2026
MAASTRICHT

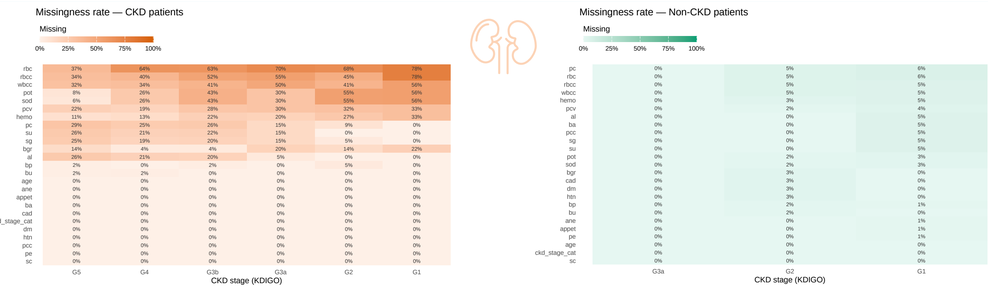
Structured Missingness in Synthetic Clinical Tabular Data: An Experimental Evaluation of Missingness Preservation, Predictive Performance, and Privacy

Clinical and Synthetic Data

- Restricted clinical data takes months to access. Synthetic data can give researchers a first look, if it preserves the real data's properties.







What properties are important?

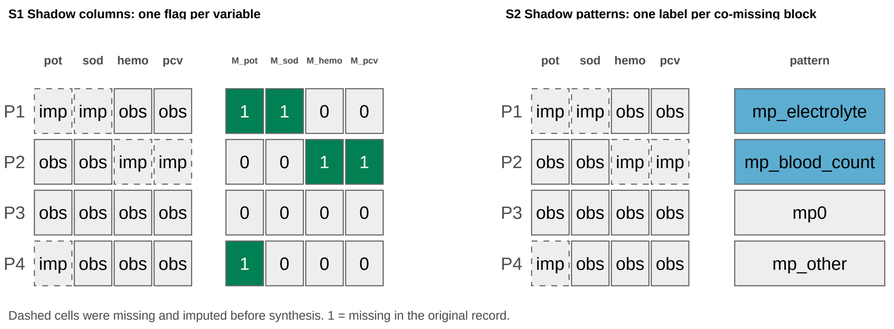
- Clinical records are rarely complete, and missingness is structured: tests tend to be missing together.

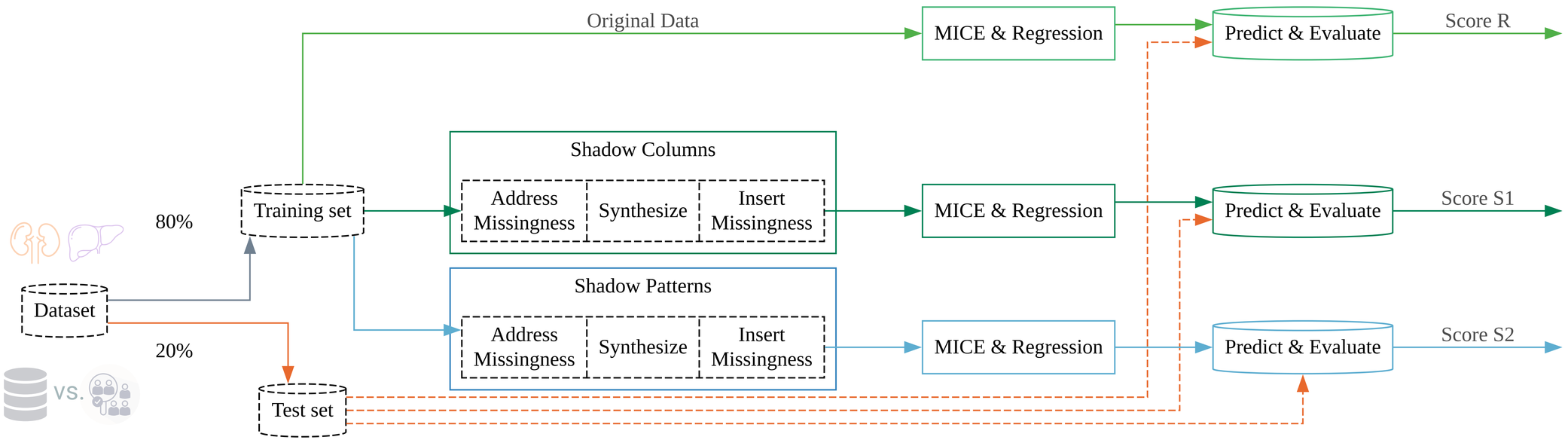
Does explicitly representing co-missingness improve synthetic data?

Main Experiment Setup

Approach

- Datasets: CKD (n = 400) and Hepatitis (n = 155), UCI
- S1 Shadow columns: missingness per variable
- S2 Shadow patterns: missingness as co-missing blocks
- Evaluated Bayesian network model with TSTR on prediction, pattern preservation and singling-out privacy risk, including subgroups

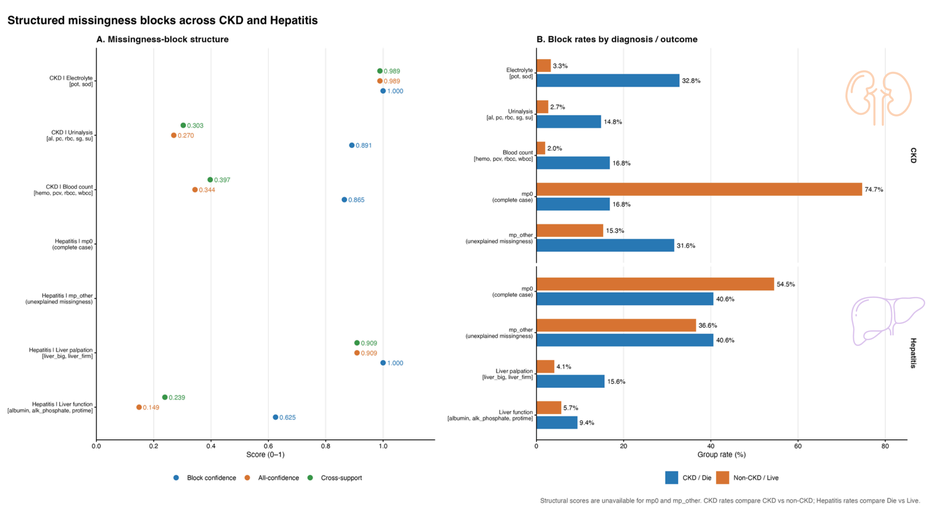




Results

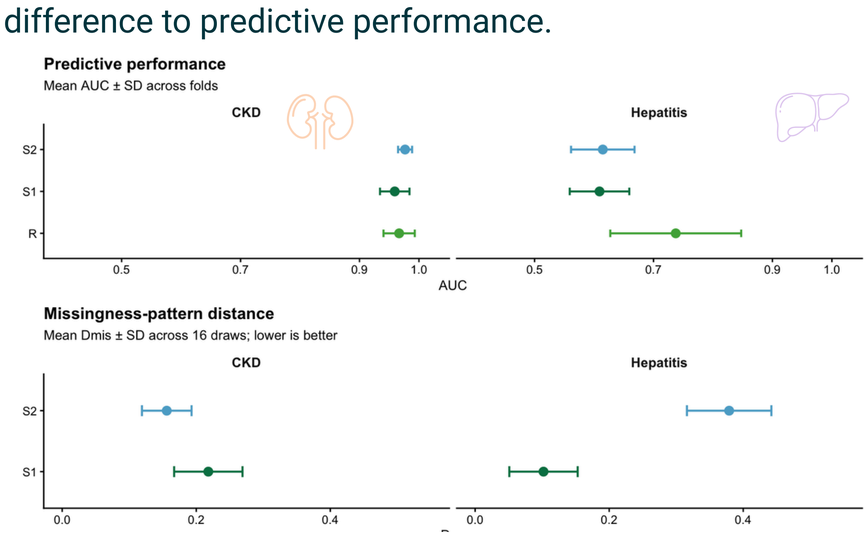
Missingness and Patterns

CKD shows more pronounced co-missingness than Hepatitis, and its blocks resemble recognizable test panels.



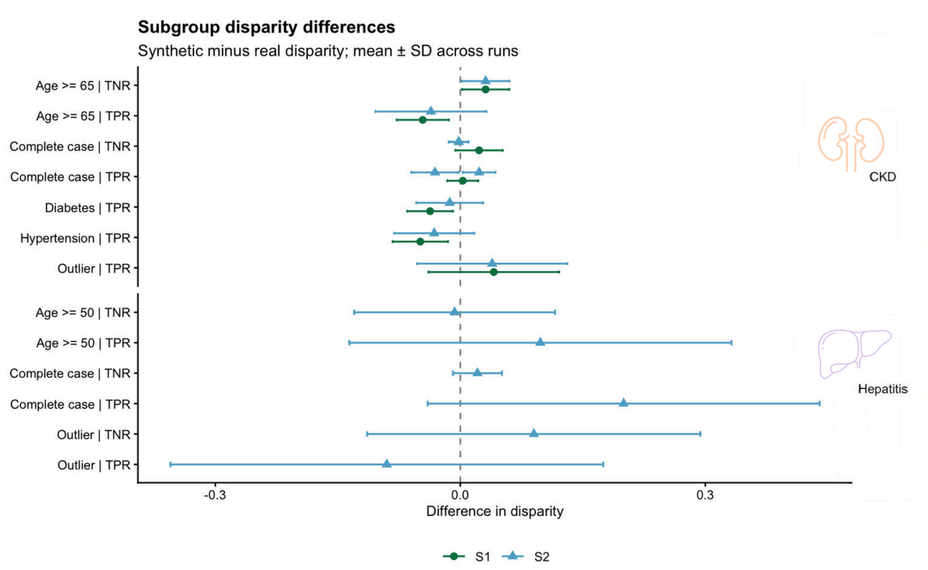
Predictive Performance

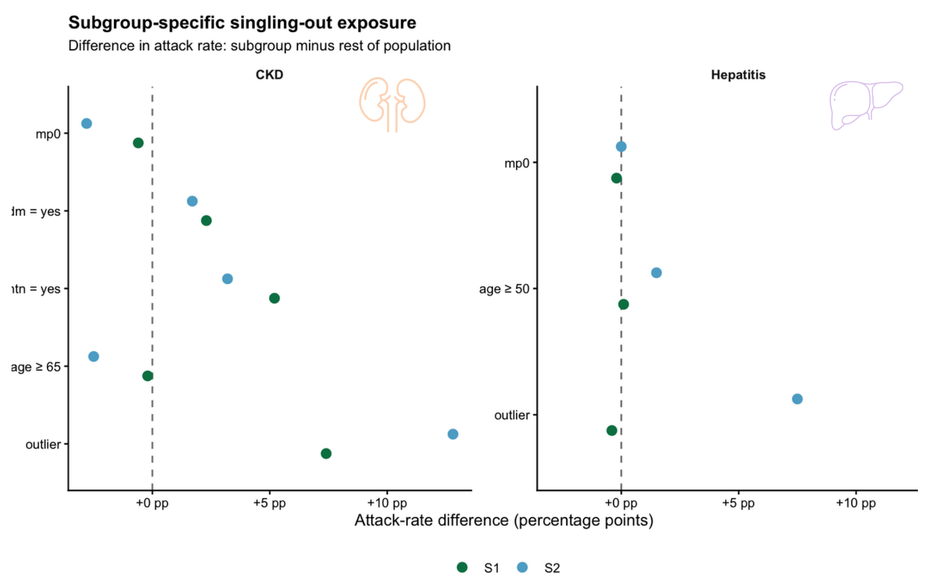
Explicitly representing co-missingness helped pattern preservation in CKD but not in Hepatitis, and made little difference to predictive performance.



Subgroups

Good overall performance does not guarantee similar behaviour across subgroups, and unusual patients (CKD outliers) were more exposed to singling-out attacks.





Limitations

- Two datasets and few synthesis runs; results are descriptive and the generator is not fully deterministic.
- Findings depend on our mining thresholds, the downstream model and the privacy attack used.

Future Work

- More datasets and synthesis runs.
- Refine the missingness mining (thresholds and block definitions).
- More robust imputation before synthesis.
- Other synthesis models and configurations.
- More metrics for fidelity, fairness and privacy.